

KRR Practical: Sorting Algorithms (Bubble + Selection Sort)

Rule: IF $A > B$ THEN Swap(A,B)

```
def bubble_sort(arr):
    for i in range(len(arr)):
        for j in range(len(arr)-i-1):
            if arr[j] > arr[j+1]:
                arr[j], arr[j+1] = arr[j+1], arr[j]
    return arr

def selection_sort(arr):
    for i in range(len(arr)):
        m=i
        for j in range(i+1,len(arr)):
            if arr[j] < arr[m]:
                m=j
        arr[i], arr[m] = arr[m], arr[i]
    return arr

numbers=[64,25,12,22,11]
print("Original:", numbers)
print("Bubble:", bubble_sort(numbers.copy()))
print("Selection:", selection_sort(numbers.copy()))
print("Complexity:  $O(n^2)$ , Space:  $O(1)$ ")
```

Output: [11,12,22,25,64]